METRIC

MIL-DTL-14929A
29 May 2018
SUPERSEDING
MIL-S-14929
August 30, 1968

DETAIL SPECIFICATION

SULFUR, COMMERCIAL GRADE

(For use in ammunition)

Reactivated after 29 May 2018 and may be used for new and existing designs and acquisitions

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for commercial grade Sulfur and is intended for use in ammunition.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: RDAR-EIQ-SA, Picatinny Arsenal, New Jersey 07806-5000 or email to usarmy.picatinny.ardec.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

AMSC N/A FSC 1375

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specification and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://quicksearch.dla.mil)

2.3 Order of precedence. Unless otherwise specified in this document or in the contract, in the event of a conflict between the text of this document and references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (See 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.1.2 Conformance. A sample shall be subjected to conformance inspection in accordance with 4.4.

3.2 Chemical requirements. Sulfur shall conform to the limits for the properties specified in Table I, when determined as specified in the applicable subparagraphs of 4.5.

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TABLE I. Chemical requirements.

Properties	Minimum percent	Maximum percent
Sulfur (dry basis)	99.5	-------
Moisture	-------	0.5
Ash	-------	0.1
Organic impurities	-------	0.1

3.3 Foreign material. There shall be no foreign contaminants including dust, debris from machinery, or any other particle that is not intended to be in the material.

3.4 Workmanship. Care shall be taken to protect the material and its components from moisture and adverse environmental effects. This may be evidenced by leakage, loss of contents, objectionable dents, or corrosion. All required markings shall be correct, neat and legible.

4. VERIFICATION

TABLE II. Requirement/verification cross reference matrix.

Method of Verification 1 - Analysis 2 - Demonstration (end item test) 3 - Examination 4 - Test		Classes of Verification A - First article inspection B - Conformance inspection						
Description	Section 3 Requirements	Section 4 Verifications	Verification Method				Verification Class	
			1	2	3	4	A	B
First article	3.1.1	4.3			X	X	X	
Conformance	3.1.2	4.4			X	X		X
Chemical requirements	3.2	4.5.2				X	X	X
Sulfur (dry basis)	3.2	4.5.2.4				X	X	X
Moisture	3.2	4.5.2.1				X	X	X
Ash	3.2	4.5.2.2				X	X	X
Organic impurities	3.2	4.5.2.2				X	X	X
Foreign material	3.3	4.4.2.1			X		X	X
Packaging	5.1	4.4.2.1			X		X	X
Workmanship	3.4	4.4.2.1			X		X	X

4.1 Classification of inspection. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Conformance inspection (see 4.4)

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4.2 Verification conditions. Unless otherwise specified, all verifications shall be performed in accordance with the test condition specified in the applicable test method document of the paragraphs in the specification.

4.3 First article inspection. When specified, a sample shall be subjected to first article inspection in accordance with Table III.

4.3.1 Inspections to be performed. The first article inspection shall be 100% examination of all characteristics including workmanship requirements for the components listed and the tests specified in Table III.

4.3.2 Rejection. The first article sample shall be rejected if any of the inspections/tests in Table III are not acceptable.

TABLE III. First article.

	No. of sample units	Requirement paragraph	Inspection method
Sulfur (dry basis)	Note 1	3.2	4.5.2.3
Moisture	Note 1	3.2	4.5.2.1
Ash	Note 1	3.2	4.5.2.2
Organic impurities	Note 1	3.2	4.5.2.2, 4.5.2.2.1
Foreign material	Note 1	3.4	4.4.2.1
Note 1. Samples shall be taken from a 1200 gram sample representing the lot in accordance with 4.5.1.			

4.4 Conformance inspection.

4.4.1 Lot formation. Inspection lots shall comply with the lot formation provisions of MIL-STD-1916. A lot shall consist of an essentially homogeneous batch of sulfur from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer in one unchanged process.

4.4.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for Critical, Major, and Minor characteristics are as defined in MIL-STD-1916. In addition to the sampling requirements stated herein, Verification Level VII shall be used to verify the 100% screening operation of Critical characteristics. Unless otherwise specified, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

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b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.4.2.1. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

4.4.2.1	Sulfur.			
Classification	Examination or test	Conformance criteria	Requirement paragraph	Inspection method reference
Critical	None defined			
Major				
101	Sulfur (dry basis)	4.5.1	3.2	4.5.2.3
102	Moisture	4.5.1	3.2	4.5.2.1
103	Ash	4.5.1	3.2	4.5.2.2
104	Organic impurities	4.5.1	3.2	4.5.2.2
105	Foreign material	Level IV	3.3	Visual
Minor				
201	Container damaged	Level II	5.1	Visual
202	Container unsealed	Level II	5.1	Visual
203	Weight markings incorrect	Level II	5.1	Visual
204	Marking missing or incorrect	Level II	5.1	Visual
205	Workmanship	Level II	3.4	Visual

4.5 Methods of inspection.

4.5.1 Sampling. A representative sample of approximately 1200 g shall be taken from each estimated tenth of the lot. The ratio of lumps to fines shall be approximately the same in the sample as in the lot. The 10 samples shall be combined to form a composite sample and handled in the following manner: crush the lumps to reduce the composite sample to laboratory size by the usual quartering process. Place the sample in a clean, dry, glass bottle and stopper tightly. Label the bottle in a manner that will identify the sample with the lot represented. If the lot does not comply with paragraph 3.2 & 3.4, the lot shall be rejected.

4.5.2 Test methods and procedures. The following test methods and procedures shall be performed.

4.5.2.1 Determination of moisture. Weigh approximately 50 g of the sample on a tared watch glass, heat for 12 hours at approximately 82 degrees C., cool the crucible for 1 hour at room temperature (20 – 22 °C) and weigh. Calculate the percent of moisture as follows:

$$\text{Percent moisture} = 100\,\frac{B-A}{B}$$

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Where:

A = weight of residue, g
B = weight of sample, g

4.5.2.2 Organic impurities and ash. Weigh approximately 15 g of the dried sample, (see 4.5.2.1), into the tared porcelain dish. Ignite the sample with a small pin flame. The sulfur, once ignited, will burn evenly and clean unless organic matter is present. If organic matter is present, a film of melted asphalt or oily matter forms over the surface of the molten sulfur and shuts off the air so that the flame from burning sulfur is extinguished. When a film of dark oily matter is first noticed, touch it lightly with the pin flame and it will usually break or char, allowing the sulfur to burn evenly. Toward the end of the burning, apply a very gentle heat to the dish and thus char the organic matter but keep the temperature well below red heat so that the organic matter is not ignited. When all the sulfur is burned off, as indicated by lack of sulfur dioxide odor, place the dish in a benchtop desiccator at room temperature (20 – 22 °C) for 1 hour and weigh. This weight, (C) represents the combined organic impurities and ash. Ignite the residue in the dish to burn off the organic impurities, cool the crucible for 1 hour at room temperature (20 – 22 °C) and weigh. This weight (D) represents the ash, and the difference between the two (C-D) represents the organic impurities. Calculate the percentage of organic impurities and of ash as follows:

$$\text{Percent organic impurities} = 100\,\frac{(C-D)}{E}$$

$$\text{Percent ash} = 100\,\frac{D}{E}$$

Where:

C = weight of combined organic impurities and ash, g
D = weight of ash, g
E = weight of sample, g

4.5.2.2.1 Organic impurities retest. If the percentage of organic impurities exceeds 0.1 percent as determined in 4.5.2.2, the percentage of sulfur contained in the residue shall be determined by the following procedure: weigh the same weight of sample as used in 4.5.2.2 into a tared porcelain dish. Ignite and burn the sample using the same procedure used in 4.5.2.2 until all the sulfur is burned off as indicated by lack of sulfur dioxide fumes. Prepare an oxidizing agent by melting 4 g of potassium hydroxide and 0.5 potassium nitrate in a large silver, ceramic or platinum crucible with the addition of a little water and cool the crucible for 1 hour at room temperature (20 – 22 °C). Transfer the residue quantitatively from the tared porcelain dish to the large silver, ceramic or platinum crucible. Heat gradually, stirring the mixtures frequently with a

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silver wire until all the organic matter is decomposed. Cool at room temperature for 20 minutes and dissolve the melt in approximately 300 mL of water. Acidify the resulting solution by dropwise addition of 0.10 M hydrochloric acid. Verify via pH paper. Heat to boiling and add 5 mL of a 10 percent (% w/w) aqueous barium chloride solution. Filter the solution through a tared Gooch crucible. Dry the precipitate in an oven for 1 hour, then in a muffle furnace to a dull red heat. Cool at room temperature in a benchtop desiccator for 1 hour and weigh. The true percentage of organic impurities may be calculated as follows:

$$\text{Percent sulfur in residue} = J = 100\ \frac{(F*0.1373)}{E}$$

$$\text{True percent organic impurities} = H - J$$

Where:

F = weight of barium sulfate precipitate, g
E = weight of sample, g
H = percent organic impurities as determined in 4.5.2.2
J = percent sulfur in residue as determined in this paragraph

4.5.2.3 Sulfur. Determine the percent of sulfur as follows:

$$\text{Percent sulfur (dry basis)} = 100 - (K+N+P)$$

Where:

K = percent organic impurities as determined in 4.5.2.2 or 4.5.2.2.1
N = percent ash as determined in 4.5.2.2
P = percent moisture as determined in 4.5.2.1

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5. PACKAGING

5.1 Packaging. The container shall be sealed and undamaged with the markings including weight present and correct (see 6.2).

6. NOTES

6.1 Intended use. Sulfur is intended for use in the manufacture of black powder.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Packaging requirements.
c. Requirements for first article samples (see 4.3).
d. Provision for submission of first article samples.
e. Requirements for submission of acceptance inspection equipment (see 6.3).

6.3 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: RDAR-EIQ-MA, Picatinny, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Equivalent test method approval. Prior approval of the contracting officer is required for use of equivalent test methods. A description of the proposed method should be submitted to the contracting officer. This description should include, but not be limited to, the procedures used, the accuracy and precision, and drawings of any special equipment required.

6.5 Definitions.

6.5.1 Lumps and course grained sulfur. Sulfur in the form of lumps and fines resulting from the mechanical breaking of large blocks of solid sulfur; shipments may contain 50% fines and lump diameters up to 20cm or more.

6.6 Hazardous. Caution should be exercised in the storing of sulfur as it is a combustible material under certain conditions.

6.7 Subject term (key word) listing.

Black powder
Lumps
Fines

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6.8 Change history.

Specification Revision	ECP Number	ERR Release Date
MIL-S-14929	-	30-Aug-1968
MIL-DTL-14929A	Y18Q2001	-

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – AR
Navy – OS
Air Force – 11

Preparing Activity:
Army – AR
(Project 1375-2018-005)

Review activities:
Army – AV, MI
Navy – AS
Air Force – 70

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